Total No. of Questions : 8]

SEAT No. :

# PE4288

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## [6582]-61

## S.E. (Information Technology)

## DATA STRUCTURES AND ALGORITHM

## (2019 Pattern) (Semester - III) (214443)

*Time : 2½ Hours]* *[Max. Marks : 70*

***Instructions to the candidates:***

1) ***Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Figures to the right indicate full marks.***
4) ***Assume suitable data, if necessary.***

***Q1)*** a) Explain Representation of stack using array, structure and linked list with example. **[6]**

b) Write a short note on doubly ended queue with example. **[6]**

c) Explain sequential organization of queue. With example. **[6]**

OR

***Q2)*** a) Evaluate the following postfix expression AB + C – BA + C \$ for A = 1, B = 2, C = 3. **[6]**

b) Convert following infix expression to postfix expression. **[6]**

(a+b)*(c/d)-e

c) Explain operations on stack with example. **[6]**

***Q3)*** a) Explain representation of tree using sequential and linked organization. **[6]**

b) Discuss the merits & demerits of implementing threaded binary tree. **[6]**

c) Explain binary tree traversals with example. **[5]**

OR

***Q4)*** a) Explain conversion of general tree to binary tree with example. **[6]**

b) Explain operations on binary search tree with example. **[6]**

c) Explain treaded binary tree with example. **[5]**

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***Q5)*** a) Explain Representation of graph with example. **[6]**

b) Write short note on prim's algorithm with example. **[6]**

c) Write short note on kruskal's algorithm with example. **[6]**

OR

***Q6)*** a) Write short note on Dijkstra's algorithm with example. **[6]**

b) Write short note on AVL tree with example. **[6]**

c) Write short note on Heap data structure with example. **[6]**

***Q7)*** a) Define hash table and Explain Basic Concepts of hashing. **[6]**

b) Explain Linear Probing with example. **[6]**

c) Explain Quadratic Probing with example. **[5]**

OR

***Q8)*** a) Explain any three operations on file with example. **[6]**

b) Differentiate between index sequential and direct access file. **[6]**

c) Explain sequential file and index sequential file organisation. **[5]**

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